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RESULTS

Results

01 · POISSON / NEGATIVE BINOMIAL

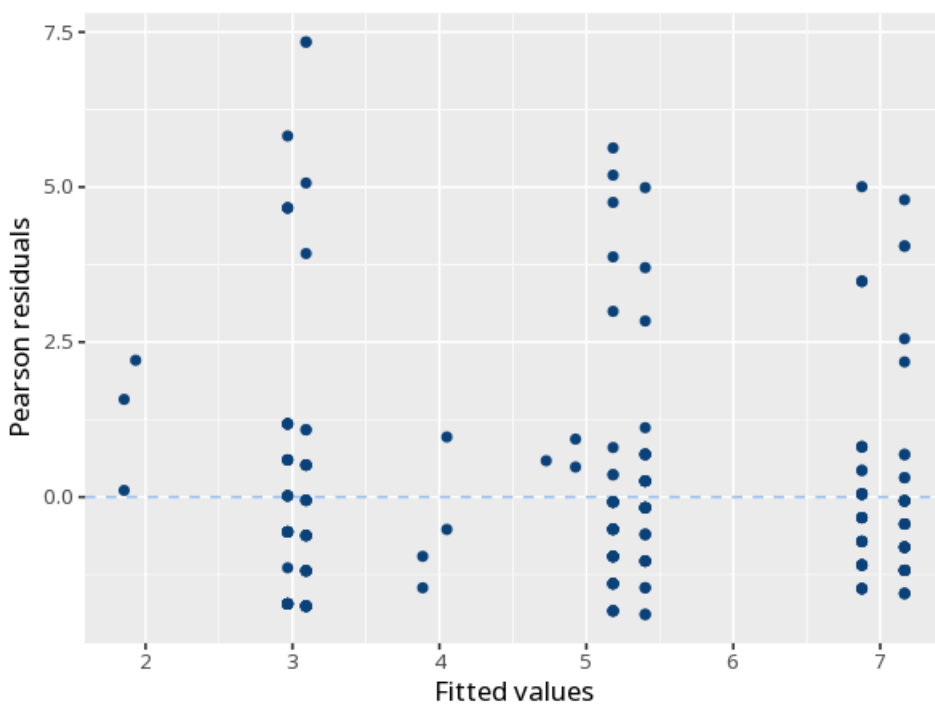
predict a count outcome

Table 1. Coefficients

	Log-count (B)	IRR
(Intercept)	−1.09 (0.06)	0.34 [0.30, 0.38]
teaching_method: Flipped	−0.56 (0.08)	0.57 [0.49, 0.67]
teaching_method: Lecture	0.28 (0.06)	1.33 [1.17, 1.51]
gender: Male	−0.04 (0.06)	0.96 [0.86, 1.07]
Num.Obs.	240	
Dispersion	3.14	
Residual deviance	633.11	
df	236	
Log.Lik.	−683.51	
AIC	1375.03	
BIC	1388.95	

dispersion check (Poisson): if variance >> mean (over-dispersion), switch to negative binomial, which instead reports theta (its overdispersion parameter) rather than this ratio. Overdispersion test: $\chi^2(236)$ test $p < .001$ — overdispersion detected.

Figure. Fit / residuals



Understanding the numbers

Incidence-rate ratio (IRR). The multiplicative change in the expected count per one-unit increase in the predictor (>1 more, <1 fewer). Here, $IRR = 0.57$.

Log-count (B). The raw coefficient on the log-count scale that the IRR is exponentiated from. Here, $B = -0.56$ on the log-count scale.

Dispersion. Checks whether the counts are over-dispersed (variance much greater than the mean) - if so, a negative-binomial model (reporting θ instead) fits better than Poisson. Here, dispersion = 3.14, overdispersion test $p < .001$.

Residual deviance. How much variation in the counts remains unexplained by the model - compare it to its degrees of freedom to gauge fit. Here, residual deviance = 633.11.

df. The degrees of freedom for the residual deviance, i.e. observations minus estimated parameters. Here, $df = 236$.

Log.Lik.. The log-likelihood: how probable the observed counts are under the fitted model; it underlies AIC and BIC. Here, Log.Lik. = -683.51.

AIC. A fit measure that penalizes model complexity, useful for comparing models fit to the same data (lower is better). Here, AIC = 1375.03.

BIC. Like AIC but penalizes complexity more heavily as sample size grows (lower is better). Here, BIC = 1388.95.

N. The number of observations the model was fit on. Here, $N = 240$.

How to read this test

Models event counts. Each predictor's incidence-rate ratio (IRR) gives the multiplicative change in the expected count per unit (>1 more, <1 fewer); p tests significance. Check dispersion to pick Poisson vs. negative binomial. If cases have

unequal exposure (different observation time/area/population), add an offset of $\log(\text{exposure})$ so the model predicts rates, not raw counts — omitting it biases the IRRs. Your significance threshold (α) is 0.05.

APA template: Predictor teaching_method: Flipped was associated with the count, $\text{IRR}=0.57$, 95% CI [0.49, 0.67], $p < .001$.

Statistical basis: Poisson regression (incidence-rate ratios) (R Core Team (2026). R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna.); Negative binomial regression (glm.nb) for overdispersed counts (Venables WN, Ripley BD (2002). Modern Applied Statistics with S, Fourth edition. Springer, New York.); Overdispersion check (Lüdecke D, Ben-Shachar MS, Patil I, Waggoner P, Makowski D (2021). "performance: An R Package for Assessment, Comparison and Testing of Statistical Models." Journal of Open Source Software, 6(60), 3139.). Full references in the exported CITATIONS.txt.

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